

# RAJIB MAHATA

Senior Software Engineer | .NET & Azure | Software Architecture | AI/GenAI Product Engineering

Kolkata, India | rajibmahata143@gmail.com | rajibmahata143@outlook.com | +91 84202 49020 | +91 91001 84730 |  
linkedin.com/in/rajib-mahata | github.com/rajibmahata | rajiblabs.com

## PROFESSIONAL SUMMARY

Senior software engineer with 10+ years designing, building, and modernizing enterprise applications in .NET, Azure PaaS, and cloud-native architecture - including seven years leading pharmacy digital-transformation systems for a Fortune 500 retail healthcare client. Skilled in microservices, CQRS, and API-first system design, with hands-on delivery of production Azure serverless and PaaS solutions. Complements enterprise engineering depth with a portfolio of independently designed and shipped AI/GenAI products - retrieval-augmented generation systems, LLM-orchestrated decision agents, and multi-tenant SaaS platforms - covering architecture, backend, frontend, and go-to-market. Comfortable owning a system end-to-end, from design documents through production deployment.

## CORE EXPERTISE

- Enterprise .NET & Azure Architecture - Microservices, CQRS, Clean Architecture, Design Patterns, SOLID
- Cloud-Native & Serverless Engineering - Azure Functions, Logic Apps, Service Bus, Event Grid
- API Design, Systems Integration & Data Engineering
- AI/GenAI Product Engineering - RAG, LLM Orchestration, AI Agents, Vector Databases
- SaaS & Multi-Tenant Platform Design
- Healthcare/Retail Domain - Pharmacy Digital Transformation, Regulated Data Handling
- Technical Specification, Solution Design & Client-Facing Delivery

## TECHNICAL SKILLS

**Languages & Frameworks:** C#, .NET 6/8/10, ASP.NET Core, ASP.NET MVC, Web API, Blazor, Entity Framework, LINQ, ADO.NET, WCF, Python (FastAPI), JavaScript/TypeScript, React, Angular, Node.js

**Architecture & Engineering:** Microservices, CQRS, Clean Architecture, SOLID, Design Patterns (Factory, Adapter, Singleton), REST API Design, System Design, UML

**Cloud & DevOps:** Microsoft Azure PaaS, Azure Functions, Logic Apps, Service Bus, Event Grid, Data Factory, App Service, Azure DevOps, Docker, Docker Compose, CI/CD, Git

**Databases:** SQL Server, PostgreSQL, Cosmos DB, Qdrant, ChromaDB, Oracle

**AI & GenAI:** LLM Orchestration (GPT-4o, DeepSeek), Retrieval-Augmented Generation (RAG), AI Agents, Vector Embeddings, Azure Document Intelligence (OCR), Speech-to-Text Pipelines, Prompt Engineering

**Product & Platform:** Multi-Tenant SaaS Architecture, Real-Time Systems (WebSocket, SignalR), Payment/GST Workflows, Product Pricing & GTM

## PROFESSIONAL EXPERIENCE

### Tata Consultancy Services - Assistant Consultant

August 2019 – Present

Client: Meijer Inc. | Kolkata, India | Pharmacy Digital Transformation

- Led development of open APIs that reduced pharmacy reliance on third-party vendors, improving data control and interoperability across internal and external systems.
- Architected a prescription automation platform - including a custom rule engine, automated SMS/voice notification pipeline, and configurable pickup workflows - reducing refill processing time by over 30% and cutting medication errors linked to refills by 40%.
- Designed and implemented an automated prescription refill and reminder system handling secure patient data in compliance with healthcare regulations, contributing to a 25% increase in patient satisfaction scores.
- Built a configurable appointment booking system for immunization programs during the pandemic, improving scheduling efficiency and service accessibility for healthcare providers.

- Delivered the pharmacy transformation program end-to-end within a 48-week engagement, deploying a rule engine and multiple Progressive Web Apps on Azure to replace vendor-dependent processes.
- Engineered Azure serverless components (Functions, Logic Apps, Service Bus, Event Grid) and Cosmos DB data pipelines to support real-time prescription and appointment processing.
- Eliminated manual IVR/phone-based refill workflows, reducing operational cost while improving the customer experience.

#### **Accenture - Software Developer**

*July 2016 – February 2019*

*Client: Cincinnati Bell, Inc. | Hyderabad, India | Communication Media Technology (CMT) Application*

- Designed, developed, and maintained the CMT application supporting network equipment provisioning for individual and business telecom customers.
- Automated legacy manual workflows in the order-fulfillment and network-configuration process, reducing manual intervention by 30% and processing time by 40%.
- Built and maintained an issue-resolution ticketing system, achieving a 95% resolution rate for user-reported issues within 24 hours.
- Redesigned the ordering system and order-fulfillment pipeline, enabling field engineers to configure network devices more efficiently.
- Improved user satisfaction scores by 25% through interface redesign and streamlined issue resolution.

#### **The Keshri Software Solutions - Web Developer**

*March 2013 – April 2016*

*Kolkata, India | Early Career*

- Designed and developed multiple ASP.NET MVC / Entity Framework web applications end-to-end across media, classifieds, transportation, and health/wellness domains, covering front-end, back-end, database design, and production deployment (see Professional Projects below).

### **PROFESSIONAL PROJECTS - CLIENT & EMPLOYER WORK**

---

#### **Pharmacy Business Transformation** - *Meijer Inc. - TCS, 2019–Present*

Reduced pharmacy dependency on third-party vendors through open APIs; deployed a rule engine and multiple Progressive Web Apps on Azure within a 48-week engagement, giving the client direct control over prescription data and processes.

#### **Smart Refilling Process (Automated Prescription Refill System)** - *Meijer Inc. - TCS, 2019–Present*

Automated manual refill workflows with a custom rule engine and multi-channel (SMS/voice) reminders; reduced refill processing time by over 30% and refill-related medication errors by 40%.

#### **Vaccine Distribution Appointment App** - *Meijer Inc. - TCS, 2019–Present*

Configurable appointment booking platform for immunization programs, streamlining pandemic-era vaccine scheduling and improving service accessibility for healthcare providers.

#### **Communication Media Technology (CMT) Application** - *Cincinnati Bell, Inc. - Accenture, 2016–2019*

Network equipment provisioning platform for individual and business telecom customers; automated legacy order-fulfillment workflows, cutting processing time by 40% and manual intervention by 30%.

#### **Cinematic Lens** - *Keshri Software Solutions - Company Product, Jul 2015 – Feb 2016*

Corporate marketing video platform with a video repository, shot cataloging, model portfolios, and a client review/feedback system.

#### **Corporate Hour** - *Keshri Software Solutions - Company Product, Jul 2014 – Feb 2015*

Online classifieds marketplace with role-based user permissions and an admin portal for listings and ad management.

#### **TRANSZOOM** - *Keshri Software Solutions - Company Product, Jul 2014 – Feb 2015*

Car rental aggregator connecting customers with transportation operators across India.

#### **Empowering Weighs** - *Keshri Software Solutions - Company Product, Jul 2013 – Feb 2014*

Medically guided weight-management platform with BMR-based calorie calculations and personalized daily suggestions.

#### **TruckIt365** - *Keshri Software Solutions - Company Product*

Freight-matching platform for shippers and carriers. Listed among your area/application highlights in the source resume with no further build detail - dates and scope are left as-is for you to confirm or expand.

## SELECTED AI / PRODUCT ENGINEERING PROJECTS - RAJIBLABS

---

*Self-directed learning projects and products built alongside full-time employment to explore AI/GenAI architecture and product engineering, under RajibLabs ([rajiblabs.com](https://rajiblabs.com)).*

### **ReturnGuard AI** - Omnichannel Retail Returns Fraud Detection Copilot

Designed and built an AI copilot that analyzes return signals across channels and outputs a four-way decision (approve / reject / partial refund / escalate) with agent- and customer-facing explanations. Architected a FastAPI backend with WebSocket support and Redis session state, a ChromaDB vector store using multilingual-e5-large embeddings, GPT-4o/GPT-4o-mini split-reasoning, a Whisper-based voice pipeline, and a React (TypeScript/Zustand/Tailwind) copilot UI, validated with Playwright end-to-end tests.

### **HistoriaAI** - Bengali-First AI History Tutoring Platform for Indian K-12

Designed a multilingual RAG tutoring platform with a human-in-the-loop content-validation pipeline (AI-generated, tutor-approved), AI-generated exam questions with per-point explanations, and a real-time tutor dashboard. Built on Python/FastAPI, PostgreSQL, and per-language ChromaDB collections with GPT-4o/GPT-4o-mini, Azure Document Intelligence OCR, and Azure Batch STT; defined the freemium pricing model (₹79/month, ₹799/year).

### **LexVault** - Legal Document RAG System

Architected a dual-pipeline legal document retrieval system pairing Qdrant hybrid semantic search with a deterministic clause rule engine to reduce hallucinated answers, built on .NET 8 and deployed on-premise on Windows Server 2022.

### **InboxPilot** - Multi-Tenant SaaS Email Triage Platform

Reframed a freelance automation proposal for a German GDPR consultancy into a productized, multi-tenant SaaS with a two-gate LLM classification architecture (n8n, Langdock, OpenRouter) to reduce false-positive triage.

### **PestFlow** - Multi-Tenant Pest Control Service Platform - In Development

Architecting a multi-tenant SaaS platform for pest control service companies, covering GST-compliant customer quotations, flexible online/cash payment plans, technician job workflows with in-app verification codes, and real-time updates via SignalR and Firebase Cloud Messaging, built on .NET and SQL Server. Repository: [github.com/rajibmahata/pestflow](https://github.com/rajibmahata/pestflow).

## EDUCATION

---

B.Tech, Computer Science and Engineering - West Bengal Board of University, Kolkata (2012)